TASK - 5

(Laptop as Server using Vs code)

Esp32 is connected to an ultrasonic sensor which measures distance to an object and the data is sent to a server here laptop acts as server Esp is connected to the wifi and using the ip address of the laptop we can send the data to the server Here we were using Vs code and also python is used in it

Work Flow

To make this work, both your laptop and your ESP32 need to be connected to the **exact same Wi-Fi network** (like your home Wi-Fi or a mobile hotspot).
[ESP32 + Ultrasonic Sensor] →(Sends data over Wi-Fi)→[Laptop IP Address]→[VS Code Python Server]

laptop will act as the "Server" (listening for data), and ESP32 will act as the "Client" (sending the data)

Step 1: For Finding The Laptop's IP Address

We need to know our laptop's address on our Wi-Fi network so the ESP32 knows where to send the data.

1. Connect laptop to Wi-Fi.
2. Open laptop's terminal/command prompt:
   - **Windows:** Press Win + R, type cmd, and press Enter. Type ipconfig and press Enter. Look for **IPv4 Address** (it usually looks like 192.168.X.X or 10.X.X.X).

Step 2: Set Up the Laptop Server in VS Code

1. Open **Visual Studio Code**.
2. Install the **Python extension** by clicking the Extensions icon on the left sidebar (the 4 squares), searching for "Python", and clicking **Install**.
3. Create a new file, name it laptop_server.py, and paste the following code:

```
import socket

# Configuration
# '0.0.0.0' tells your laptop to listen to any device on the network
SERVER_HOST = '0.0.0.0'
SERVER_PORT = 5001          # A free port number for communication

# Create a socket object
server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```python
# Bind the socket to the host and port
server_socket.bind((SERVER_HOST, SERVER_PORT))

# Start listening for incoming connections (max 5 queued connections)
server_socket.listen(5)
print(f"[*] Server is listening on port {SERVER_PORT}... Waiting for
ESP32...")

try:
    while True:
        # Establish connection with ESP32
        client_socket, client_address = server_socket.accept()
        print(f"[*] Connected to ESP32 at {client_address}")

        while True:
            # Receive data from the ESP32 (buffer size 1024 bytes)
            data = client_socket.recv(1024)
            if not data:
                print("[*] ESP32 disconnected.")
                break

            # Decode the bytes into a readable string
            distance_reading = data.decode('utf-8')
            print(f"Received Distance: {distance_reading} cm")

        client_socket.close()

except KeyboardInterrupt:
    print("\n[*] Shutting down server.")
    server_socket.close()
```

Step 3: Set Up the ESP32 in Thonny (MicroPython)

Connection of HC-SR04 Ultrasonic Sensor

- **VCC** to ESP32 **VIN** (or 5V / 3.3V depending on your sensor variant)
- **GND** to ESP32 **GND**
- **Trig** to ESP32 **GPIO 5**
- **Echo** to ESP32 **GPIO 18**

Open **Thonny**, create a new file, and paste this MicroPython code. **change the Wi-Fi details andtheLaptopIPaddress in thecode**

```python
import machine
import time
import network
import socket
```

```python
# 1. Hardware Pin Setup
trigger = machine.Pin(5, machine.Pin.OUT)
echo = machine.Pin(18, machine.Pin.IN)

# 2. Wi-Fi and Server Network Setup
WIFI_SSID = "YOUR_WIFI_NAME"        # Replace with your Wi-Fi Name
WIFI_PASSWORD = "YOUR_WIFI_PASSWORD" # Replace with your Wi-Fi
Password
LAPTOP_IP = "192.168.X.X"           # REPLACE WITH YOUR LAPTOP'S IP
ADDRESS
PORT = 5001

def get_distance():
    # Trigger the ultrasonic pulse
    trigger.value(0)
    time.sleep_us(2)
    trigger.value(1)
    time.sleep_us(10)
    trigger.value(0)

    # Measure the duration of the echo pulse
    while echo.value() == 0:
        signaloff = time.ticks_us()
    while echo.value() == 1:
        signalon = time.ticks_us()

    timepassed = signalon - signaloff
    # Calculate distance in centimeters
    distance = (timepassed * 0.0343) / 2
    return round(distance, 2)

# Connect to Wi-Fi
wlan = network.WLAN(network.STA_IF)
wlan.active(True)
wlan.connect(WIFI_SSID, WIFI_PASSWORD)

print("Connecting to Wi-Fi...")
while not wlan.isconnected():
    time.sleep(1)

print("Connected to Wi-Fi! ESP32 IP:", wlan.ifconfig()[0])

# Connect to the Laptop Server via Sockets
print("Connecting to Laptop Server...")
s = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
s.connect((LAPTOP_IP, PORT))
print("Connected to Server successfully!")
```

```
# Main Loop: Read sensor and send data every 2 seconds
try:
    while True:
        dist = get_distance()
        print("Local reading:", dist, "cm")

        # Convert distance to a string, then to bytes, and send it
        data_str = str(dist)
        s.send(data_str.encode('utf-8'))

        time.sleep(2)

except Exception as e:
    print("Error occurred:", e)
    s.close()
```

Step 4: How to Run and Test It

1. **Run the Laptop Server First:** Go to VS Code, right-click anywhere inside the laptop_server.py file, and select **"Run Python File in Terminal"**. You should see it print: [*] Server is listening on port 5001... Waiting for ESP32...
2. **Run the ESP32 Code:** Go to Thonny and click the green **Run** button to execute the MicroPython script on your ESP32.
3. In **VS Code Terminal** we can see it say Connected to ESP32 and immediately start printing the live distance readings every 2 seconds!